



## IT582: Foundations of Machine Learning

### Lab 2: Numerical Differentiation and Gradient Descent

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## Submission Instructions

Upload **only one Jupyter Notebook (.ipynb)** file to Moodle.  
The notebook should contain all code, outputs, plots and conclusions.  
Name the notebook as:

`[studentid]_lab2.ipynb`

Do not use automatic differentiation libraries for computing gradients or Hessians.

## Task 1: Numerical Gradient

For a scalar-valued function

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

the gradient of  $f$  at  $\mathbf{x}$  is defined as

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

Implement a general function to approximate the gradient numerically using the forward finite-difference formula:

$$\frac{\partial f(\mathbf{x})}{\partial x_i} \approx \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h},$$

where  $h > 0$  is a small step size and  $\mathbf{e}_i$  is the  $i^{\text{th}}$  standard basis vector.  
Use the following functions:

$$f_1(x) = 2x^2 + 3x + 1$$

and

$$f_2(x, y) = 2x^2 + xy + 3y^2 - 4x + 2y + 5$$

Compute the numerical gradient at suitable points.

## Task 2: Numerical Hessian

For a scalar-valued function

$$f : \mathbb{R}^n \rightarrow \mathbb{R},$$

the gradient is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

The Hessian is obtained by differentiating the gradient with respect to the input variables:

$$H(\mathbf{x}) = \nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

Using the numerical gradient implemented in Task 1, approximate the  $i^{\text{th}}$  column of the Hessian as

$$H_{:,i}(\mathbf{x}) \approx \frac{\nabla f(\mathbf{x} + h\mathbf{e}_i) - \nabla f(\mathbf{x})}{h},$$

where  $\mathbf{e}_i$  is the  $i^{\text{th}}$  standard basis vector.

In the above expression, both gradient terms must be evaluated using the **numerical gradient function developed in Task 1**.

Implement a general function:

```
numerical_hessian(f, x, h)
```

by performing the following steps:

1. Compute the numerical gradient at  $\mathbf{x}$ .
2. Perturb one input variable at a time by  $h$ .
3. Recompute the numerical gradient at the perturbed point.
4. Use the change in gradient to obtain the corresponding column of the Hessian.

Calculate the numerical Hessian for both  $f_1$  and  $f_2$  at the same points used in Task 1.

## Task 3: Validation with Analytical Derivatives

For  $f_1$  and  $f_2$ :

1. Derive and implement the analytical gradient.
2. Derive and implement the analytical Hessian.
3. Compare analytical and numerical gradients.
4. Compare analytical and numerical Hessians.
5. Report the absolute error between the two results.

## Task 4: Numerical Gradient for Linear Regression

Use the **California Housing Dataset** and consider only:

- Input feature: `MedInc`
- Target: `MedHouseValue`

Use the same data splitting and standardization procedure followed in Lab 1.

The Linear Regression model is:

$$\hat{y} = \theta_0 + \theta_1 x$$

with Mean Squared Error:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m (\hat{y}_i - y_i)^2$$

Implement the gradient of the MSE using:

- Analytical differentiation
- Numerical differentiation

Use the same initialization, learning rate and number of iterations for both methods. Compare gradients and report the difference.

## Task 5: Batch, Mini-Batch and Stochastic Gradient Descent

Using the Linear Regression model from Task 4, implement:

- Batch Gradient Descent
- Mini-Batch Gradient Descent
- Stochastic Gradient Descent

Run each method using:

- Analytical Gradient
- Numerical Gradient

Plot:

- Epoch vs Training MSE
- Epoch vs Validation MSE
- Bar plot comparing execution time

Write conclusions comparing:

- Analytical and numerical gradients.
- Batch, Mini-Batch and Stochastic Gradient Descent.
- Convergence behaviour.